Preme Malaviphusit

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EDUCATION

New York University, Tandon School of Engineering – New York, NY — Expected May 2028

Bachelor of Science in Electrical and Computer Engineering | Mathematics Minor | GPA: 3.73

- Related Coursework: PCB for Embedded System, Digital Logic, Circuits, Electronics, Data Structure and Algorithms

EXPERIENCE

NYU General Engineering Department – Brooklyn, NY — Aug 2025 - Present

Head of Prototyping, Teaching Assistant

- Managed Prototyping Lab for 400+ students and trained TAs in embedded systems, soldering, CAD, and 3D printing
- Led hands-on labs for 16 students covering circuit design, firmware development, CAD, and sensor data acquisition
- Mentored 4 project teams through full hardware-software integration, resulting in 100% on-time prototype delivery

Flexible AI-Enabled Mechatronic Systems Lab – Brooklyn, NY — Sep 2025 - Dec 2025

Computer Vision Researcher

- Developed a kidney stone detection pipeline using TensorFlow and PyTorch for real-time endoscopic imaging
- Generated 3D volumetric heat maps in slicer software to support laser ablation planning

Metropolitan Electricity Authority – Bangkok, Thailand — Jun 2023 - Jul 2023

Electrical Engineering Intern

- Developed electrical system documentation and load analysis for the Thai Parliament substation using AutoCAD
- Automated signal transmission analysis in Excel, cutting manual design review time by 90%
- Inspected 5+ electrical systems across solar, EV, and hospital infrastructure for performance and compliance

TECHNICAL PROJECTS

SharkFin Rev 1.0 Avionics Board | NYU ARC Robotics | Altium Designer, LTSpice — Jan 2026 - Present

- Designed a 4-layer STM32H743 avionics board for guided projectile stabilization with IMU and OV5460 camera
- Implemented dual-input USB-C/LiPo power mux with reverse-polarity protection and 0.8 mm BGA fanout

Payload Electronics & RF Telemetry | NYU Rogue Aerospace | EasyMini, Altus Metrum, XBee — Jan 2026 - Present

- Engineered flight-ready autonomous payload PCB for NYU's first hybrid rocket, ejecting water at apogee (~1,900 ft)
- Integrated XBee RF telemetry to downlink tank pressure, altitude, and GPS coordinates to ground station

Drone Flight Controller | Altium Designer — Apr 2026 - May 2026

- Designed STM32F405 flight controller PCB with 4-channel PWM ESC outputs, iBUS receiver, and SWD interface
- Implemented 5–36V wide-input power architecture supporting 2S-8S LiPo with onboard MCU/peripheral regulation

Custom STM32F405 Development Board | Altium Designer — Jan 2026 - Mar 2026

- Developed a 4-layer STM32F405 PCB exposing SPI, I2C, UART, and ADC peripherals with USB-C ESD protection
- Architected 36V XT30 to 5V buck and 3.3V LDO power for stable MCU operation under variable load
- Integrated SWD debug interface with differential pair routing to ensure high-speed signal integrity

LEADERSHIP / EXTRACURRICULAR

Secretary, NYU Society of Asian Scientists and Engineers — May 2026 - Present

Vice President, NYU Society of Women Engineers | Outstanding Event Management Award — Oct 2025 - Present

President, NYU Thai Student Association | Represented NYU at 6,000+ attendee university fair — Jul 2025 - Present

SKILLS

Hardware/PCB: Altium, KiCad, LTSpice, ModelSim, Vivado, Soldering, Oscilloscope, Multimeter, Power Supplies

Embedded Systems: Arduino, ESP32, STM32, Raspberry Pi, SPI, I2C, UART

Programming: C/C++, Python, MATLAB, OpenCV, Verilog, Java, SQL, HTML/CSS, Git

Mechanical/CAD: Fusion360, SolidWorks, OnShape, AutoCAD